Geocomputation & Machine Learning course

Quick recap

The meeting focused on a recap and hands-on session regarding optimizers and the perceptron for tree height prediction. Raffaele reviewed the basics of linear regression, the role of bias, and the use of MSE loss, explaining that the closed-form solution is computationally unfeasible for large datasets. He discussed regularization to smooth the loss landscape and prevent overfitting, and explained how activation functions like sigmoid, ReLU, and tanh scale outputs for classification or regression tasks. The group then performed a hands-on tutorial using a tree height dataset, where they compared random forest predictions with a perceptron and observed how different learning rates affected convergence and R-squared values. Raffaele explained the limitations of stochastic gradient descent, such as getting stuck in local minima, and introduced mini-batch training and the concept of momentum to improve optimization. He further compared SGD, Adagrad, and Adam optimizers, illustrating how Adam uses adaptive learning rates and momentum to converge more efficiently toward the global minimum. The session concluded with a discussion on early stopping to prevent overfitting and a brief introduction to feed-forward neural networks for extracting more complex patterns.

Next steps

Raffaele

- Start the next lecture on Tuesday with a hands-on session using the Adam optimizer on the tree height dataset.
- Introduce feed-forward neural networks in the next lecture to enable extraction of more complex patterns from the data.

Summary

Perceptrons and Optimization Lecture

Raffaele led a lecture on perceptrons and optimization, explaining the basics of linear combinations and how weights represent feature correlations with the target variable. He outlined the process of training models using gradient-based optimization to minimize loss functions like mean square error. The session included a hands-on tutorial on fitting a perceptron using the 3l dataset, with plans to potentially introduce feed-forward neural networks if time permitted.

Machine Learning Weight Vector Optimization

Sebastián explained the concept of optimizing weight vectors in machine learning models, discussing both vectorial and matrix formulations of loss functions. He highlighted that while the theoretical solution exists, computing it directly is impractical for large datasets due to computational constraints. Sebastián also touched on regularization as a method to address optimization issues, including the problem of local minima in the loss landscape.

Machine Learning Regularization Explained

Raffaele explained the purpose of regularization in machine learning, focusing on two main reasons: preventing model weights from getting stuck in local minima and addressing practical problems during inference. He illustrated how regularization helps distribute weight importance across features rather than relying on a single noisy feature, making predictions more robust when sensors or features produce inconsistent data. Ali asked questions about how regularization affects variable weights and whether it equally distributes importance across all features, which Raffaele clarified depends on feature correlation and scale, emphasizing that regularization aims to ensure all features contribute meaningfully to predictions.

Regularization and Data Scaling Concepts

Raffaele explained the relationship between regularization and data scaling, emphasizing that regularization must work closely with data scaling to ensure model weights are properly constrained. He discussed how regularization and activation functions work differently, with regularization controlling model weights during optimization while activation functions transform outputs to desired ranges. Raffaele also covered optimization concepts, explaining that the goal is to find the global minimum of the loss function by taking steps in the opposite direction of the gradient, and introduced the concept of learning rate alpha to control the magnitude of optimization steps.

Gradient Descent and Data Preprocessing

Raffaele explained the importance of learning rate in gradient descent optimization, noting that large learning rates can cause the model to miss the global minimum while small rates may require excessive computational steps without guaranteed convergence. The group took a break after the theoretical discussion and then began a hands-on session where Raffaele guided them through loading and preprocessing a tree height dataset, including feature selection and data scaling. The session ended with Raffaele asking the group to evaluate an R-squared plot and share their initial considerations about its quality.

Random Forest Prediction Issues Discussion

Raffaele discussed issues with random forest predictions, noting discrepancies between predicted and actual values due to unscaled data and non-Gaussian distributions. He explained the need to exclude certain variables like ID and height to avoid bias, and suggested using quantile transformation to smooth the data distribution. The team then discussed scaling the data using min-max scaling and converting it to float tensors for use with Torch, before defining a perceptron model with a linear layer.

Perceptron Model Training Discussion

Raffaele explained the structure of their perceptron model, which takes 20 features as input and predicts a single scalar value for regression tasks. He outlined the training process using different learning rates ranging from 0.5 to 10^-3, with 5,000 epochs and storing both training and validation losses. The team observed that with a learning rate of 0.5, the model never converged, as indicated by the R-square values reaching 10^33.

Model Performance Optimization Discussion

Raffaele and the team discussed the impact of learning rates on model performance, noting that a smaller learning rate improved convergence but still did not achieve optimal results. They explored the use of activation functions and identified issues with the current optimizer, suggesting a need to switch to a more effective one. Raffaele explained the concept of mini-batch training and its benefits in handling large datasets, emphasizing the importance of shuffling and sampling data to improve model performance.

Optimization Algorithms and Momentum

Raffaele explained the concept of momentum in optimization, describing how it helps address issues with gradient-based methods like SGD by incorporating a path gradient component. He also introduced Adagrad as a solution to the problem of different learning rates for different parameters, explaining how it adapts the learning rate based on accumulated gradients. The discussion concluded with a brief overview of Adam optimizer, which combines both momentum and adaptive learning rates, and Raffaele announced that they would begin implementing different optimizers and move on to feed-forward neural networks in the next lecture.